

Tax Pot Episodes in OECD Countries¹

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How to use an unexpected increase in tax revenues (tax pots) was an important issue in most OECD countries in the second half of the 1990s, the question being more precisely what to do with those windfall revenues: decrease taxes, reduce debt, increase expenditures? In this paper, we study such tax pot episodes in OECD countries over the past 40 years. To that end, we propose a definition of a fiscal pot episode. Once identification is complete, we examine the macroeconomic environment within in those episodes, the way this surplus of revenues has been used, and the degree of success in reducing public debt and in fostering growth. As in the fiscal adjustment literature, we then obtain relatively orthodox conclusions about the use of windfall tax revenues, as it is generally better for future growth and debt level to use the money to reduce expenditures and taxes. *J. Japan. Int. Econ.*, December 2002, 16(4), pp. 436–461. University of Toulouse (IDEI), Toulouse, France; and University of Toulouse (GREMAQ, LEERNA, and IDEI), Toulouse, France, Institut Universitaire de France, and CEPR. © 2002 Elsevier Science (USA)

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1. INTRODUCTION

This paper proposes descriptive analysis of tax pot episodes in OECD nations in the past 40 years. By tax pot, we mean an unexpected and non deliberated increase in tax revenues.

The question of the use of unexpected tax revenues was particularly relevant in France in 1999 and 2000. One question that arose was about the size of those *cagnottes* (French word for pot), and the exact size is still a debatable object. The

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French newspaper *Le Monde* first gave an evaluation of 66 billion French francs (February 5, 2000), which amounts to roughly .7% of GDP. It seems now that this was higher than the real size of the tax pot, which the French government eventually estimated to be 35 billion French francs. Of course, the government announcement was a strategic one, as confessing to have some extra money opens the door to new revendication (tax cuts, more subsidies, transfers, wages, etc.). During the fiscal year 2000, France again experienced an unexpected increase in tax revenues, presumably of a larger amount than the 1999 pot. The question was of course what to do with the money. Reduce public debt, increase expenditures, reduce taxes? Which taxes? Which expenditures? The French government decided to slightly reduce personal income tax and property taxes on housing and decided to abolish the automobile *vignette* (a yearly lump sum tax on cars), which was extremely unpopular. Our understanding of those decisions is that they were driven by electoral motives rather than economic ones. The present paper aims to provide guidance for the use of tax pot revenues by looking at historical episodes of tax pots.

When a government decides to follow a fiscal adjustment policy, there are some lessons that can be drawn from the past. Starting with Giavazzi and Pagano (1990), Alesina and Perotti (1995), and Alesina and Ardagna (1998) (see also Ardagna, 1999), an important literature has studied episodes of large and discretionary reductions in primary deficits. The main lessons from this literature are the following. Adjustments can be achieved in two very different ways: by increasing taxes or by decreasing expenditures. During those episodes, public investment and subsidies are the most reduced expenditures, while personal and corporate income taxes are the most increased taxes. A successful reduction of public debt within 3 years is obtained, independent of the adjustment size, by reducing expenditures, not by increasing taxes. Decreasing wage consumption and transfers increases the probability of reducing public debt, as opposed to decreasing public investment.

Here we ask whether such relatively orthodox lessons also apply to the use of tax pots. We shall use a panel of OECD countries over the past 40 years and will discuss successively the following points: (a) How to identify tax pot episodes, (b) when and where they happened, (c) the macroeconomic situation and where the pot came from, and (d) what was done with the money. Was public debt reduced? Was growth enhanced?

The paper is organized as follows. Section 2 proposes a definition and an identification method of tax pot episodes. Section 3 answers the question where and when and Section 4 answers how and to what end. Section 5 concludes, while a data description is provided in the Appendix.

2. DEFINITION AND IDENTIFICATION OF TAX POT EPISODES

2.1. Theoretical Definition and Discussion

We first propose a definition of a tax pot episode and then discuss the measurement issues related to this definition.

DEFINITION. A tax pot episode is defined as a year N in a country I during which tax revenues happened to be surprisingly and substantially large, absent any discretionary tax increase in year N .

At least three words deserve comment in this definition, “surprisingly,” “discretionary,” and to a lesser extent “substantially.” By surprisingly, we mean that the increase in tax revenues was not expected conditionally on the information available at year $N - 1$. Of course, the identification of a surprise depends on the model that is used to forecast tax revenues. By nondiscretionary, we mean that the increase of the tax revenues in year N (relative to the expectation formed in period $N - 1$) is not the consequence of a policy decision taken in period N , such as, for example, a change in a marginal tax rate. The increase in the tax revenues must be the consequence of a surprise in the size of the tax base, not a surprise in the tax schedule. What about “substantially”? Given the difficulty in identifying those surprising and nondiscretionary movements, we want to find a robust measurement tool for those episodes, i.e., a tool that selects more or less the same episodes, whatever forecasting model is used. Restricting ourselves to relatively large surprises is a way to achieve robustness.

Such difficulty in applying a definition of a relevant episode to the data is not uncommon in economics, and we shall build upon the literature on fiscal adjustment. In the fiscal adjustment literature,³ the problem is to identify deliberated reductions of the primary deficit, net of the variations related to the action of automatic stabilizers. Different solutions have been proposed to decompose primary deficit movements into exogenous policy shifts and automatic reaction of fiscal variables to the state of the economy. The OECD computes the fiscal impulse as the difference between the actual primary deficit and that which would have prevailed had expenditures increased as potential output. The International Monetary Fund (IMF) makes a similar adjustment to compute a so-called cyclically adjusted fiscal balance, but relates the current state of fiscal balance not to the previous one, but to a year in which the economy was close to its potential output. A somewhat simpler and more popular decomposition has been proposed by Blanchard (1993b): the fiscal impulse is the difference between the primary surplus in N and that which would have prevailed had unemployment stayed at its $N - 1$ level. The merit of this approach is that no potential output computation or basis year choice is needed. Nevertheless, the measure is conditional on the choice or estimation of the elasticity of the primary surplus to unemployment changes.

In this paper, we will follow a method somewhat similar to Blanchard’s to identify tax pot episodes.

³ There exists a vast literature on this topic that started with the work of Giavazzi and Pagano (1990) and Alesina and Perotti (1995), and was further extended by Giavazzi and Pagano (1995), Alesina and Ardagna (1998), Giavazzi, Jappelli, and Pagano (1998), and Ardagna (1999), among others.

2.2. Identification

Let us consider one country and assume that one can obtain a series of total tax revenues (in real terms) $\{T_t\}$, from which one can compute a time first difference $\Delta T_t = (T_t - T_{t-1})$.

A naive approach would consist in identifying a tax pot episode as a country-year in which the ratio $\Delta T_t / Y_t$ is large, where Y stands for GDP. As we mentioned before, such a measure will catch predicted, deliberated, and surprising movements in $\Delta T_t / Y_t$.

To measure the nondeliberated tax surprises, we first assume that there exists a stable linear relation between tax revenues and GDP, possibly augmented with a trend

$$T_t = \alpha_0 + \alpha_1 \Theta_t + \alpha_2 Y_t + \varepsilon_t, \quad (1)$$

where Θ is a trend between 1960 and 2000. In Eq. (1), any deliberated action taken by the tax authorities in period t is included in ε_t .

Let us assume that we can compute $E_{t-1}[Y_t]$, the conditional expectation of GDP given information available at period $t-1$ and a model for GDP. We can then construct a variable T_t^* that measures the expected tax revenues conditional on $t-1$:

$$T_t^* = \alpha_0 + \alpha_1 \Theta_t + \alpha_2 E_{t-1}[Y_t]. \quad (2)$$

The unexpected and nondeliberated variation in tax revenues will be then given by $(T_t - T_t^* - \varepsilon_t)$. A country-year in which $(T_t - T_t^* - \varepsilon_t) / Y_t$ is larger than a given threshold $\bar{\tau}$ will be considered as a tax pot episode.

Note that the total variation of tax revenues can be decomposed into three terms:

$$\Delta T_t = (T_t - T_t^* - \varepsilon_t) + (T_t^* - T_{t-1}) + \varepsilon_t. \quad (3)$$

In Eq. (3), the first term in parentheses represents the tax pot component, the second term is the expected variation in tax revenues and the unexpected variation in tax revenues that is not explained by unexpected changes in GDP. Using OECD terminology, this might be interpreted as a deliberated tax policy of period t . Because such an interpretation is fragile, as ε is also everything that is not explained by the simple model that we have used, we shall not pursue this interpretation any further.

The last point concerns the computation of $E_{t-1}[Y_t]$. We assume that the growth rate of output follows a stable $AR(p)$, possibly with a break in the constant term, that we estimate over the largest sample available for each country. Assume for simplicity that the order of the AR is one ($p=1$).

$$\Delta \log Y_t = \gamma_0 + \gamma_1 \hat{\theta}_t + \gamma_2 \Delta \log Y_{t-1} + v_t, \quad (4)$$

where $\hat{\theta}_t$ is a dummy for the subperiod 1974–2000. We then use Eq. (4) to construct a series of expected growth rates

$$E_{t-1}[\Delta \log Y_t] = \gamma_0 + \gamma_1 \hat{\theta}_t + \gamma_2 \Delta \log Y_{t-1} \quad (5)$$

and expected GDP is given by

$$E_{t-1}[Y_t] = Y_{t-1} \times (1 + E_{t-1}[\Delta \log Y_t]). \quad (6)$$

2.3. Robustness

As it is clear from the previous section, our measure is a conditional one, and it is important to judge the robustness of the episode's selection to a change in the set of assumptions under which the selection is made. We see basically four important sets of assumptions in our framework that concern respectively the data availability, the GDP forecasting model, the tax schedule, and the threshold $\bar{\tau}$. We discuss those four issues in turn.

Let us consider first data availability, and let us focus on GDP prediction (the same reasoning would also apply to the estimation of the tax schedule). Assume for simplicity of exposition that the true model for GDP is an AR(1) in growth rates. In what we have done, it is assumed that the model is known by the government and by the econometrician that makes the GDP forecast. We have therefore taken the largest possible sample to estimate that AR(1). In real life, it is likely that the model is misspecified. As a first approximation, we assume away this eventuality. It is still the case that the parameters of the model are not perfectly known by the government and that the government is learning the model in real time. In such a case, the forecast that we make today of the 1970 GDP conditional on the 1969 level of GDP given a model estimated on 1960–2000 is different from the one that was made given a model estimated on 1960–1969. A first robustness test would then recursively estimate the GDP forecasting model and use only the available information of the time to estimate the AR equation (the same exercise could also be done for the tax schedule equation). A second problem regarding data availability is related to the revision of national accounts. In year N , the GDP evaluation on $N - 1$ is not definitive and will be revised at least in N and $N + 1$. Seriously tackling the data availability problem would require the use of real time data, which are difficult to obtain for a large panel of OECD countries over a long period. For that reason, we will restrict ourselves to the use of a forecasting model estimated on the largest sample available for each country.

The second set of assumptions concerns the parametric form of the GDP forecasting model. In the baseline case presented above, we use an AR(p) on GDP growth rates. Such a univariate representation might disregard important information contained in other macroeconomic variables, for example the composition of aggregate demand or the level of inflation. Therefore, in the implementation of the episodes selection, we will also consider a more general VAR model

including GDP, nominal interest rate, consumption to GDP ratio, and the GDP deflator.

The third robustness test concerns the tax schedule, which has been assumed to be linear and including only current income. Linearity might be true only as a first approximation. For personal income tax, we know that much of the national tax systems are progressive at the individual level. But with tax brackets being continuously adjusted, it is not clear that progressivity shows up at the aggregate level. It also is unclear that the linear approximation is bad for indirect taxes such as VAT or for corporate income taxes. We will then test the robustness of the episode selection to a modification of the tax schedule that includes an additional term, the yearly GDP growth according to

$$T_t = \alpha_0 + \alpha_1 \Theta_t + \alpha_2 Y_t + \alpha_3 \Delta Y_t + \varepsilon_t. \quad (7)$$

If taxes as a whole are progressive, then α_3 will be positive and significant. In such a case, an equation that will omit the variable ΔY_t will overestimate the number of tax pot episodes. Another potential problem comes from the fact that, in some countries, some taxes are paid in period t based upon the income of period $t - 1$. It might be therefore useful to include past GDP in the tax equation and to estimate

$$T_t = \alpha_0 + \alpha_1 \Theta_t + \alpha_2 Y_t + \alpha_3 Y_{t-1} + \varepsilon_t. \quad (8)$$

We will check the robustness of our results to such a specification.

Fourth, it might be likely that a country with a high level of tax revenues hit the threshold more often than a low-level country, if the threshold is expressed in % of GDP. To correct for such a potential bias, we will also consider the following definition of a tax pot: an episode in which tax revenues/GDP surprise is higher than the country average tax surprise by an amount equal to κ times the country standard deviation of the tax revenues to GDP ratio. We will refer to this case as the relative threshold definition, as opposed to the absolute threshold definition given above.

In the next section, we implement our measure of tax pot episodes on a panel of OECD countries and conduct various robustness checks.

3. WHERE AND WHEN?

3.1. Episodes Selection

First approach. We use in this study the 1999 OECD Economic Outlook database (OCDE, 1999a) and the 1999 OECD Revenues Statistics database (OCDE, 1999b). The sample is *a priori* 1960–2000 and 23 countries will be considered. All data are annual. The list of those countries and the acronyms we use are given in Table I. In this table, we also report for each country the largest sample for which we can obtain the GDP and the total tax revenues series. In practice, we always stop in 1998.

TABLE I
List of Countries, Mnemonics, and Sample Period

Australia	AU	1962–1998
Austria	AS	1966–1998
Belgium	BE	1962–1998
Canada	CA	1963–1998
Denmark	DK	1972–1996
Finland	FI	1972–1998
France	FR	1965–1998
Germany	GE	1962–1998
Greece	GR	1962–1998
Iceland	IC	1983–1998
Ireland	IR	1979–1998
Italy	IT	1962–1998
Japan	JP	1962–1998
Korea	KO	1972–1998
Netherlands	NL	1971–1998
New Zealand	NZ	1988–1998
Norway	NO	1964–1998
Portugal	PO	1962–1998
Spain	SP	1966–1998
Sweden	SW	1962–1998
Switzerland	CH	1962–1998
United Kingdom	UK	1967–1998
United States	US	1962–1998

Note. The samples reported are those over which we have been able to estimate both the AR and VAR model for GDP and over which we have been able to compute a series of total public revenues and estimate a tax schedule.

For GDP prediction, we use $AR(p)$ models in growth rates with a dummy for post-1973 years. Using LM tests, we choose one lag for the autoregressive part in all countries except United Kingdom, Sweden, and Finland where two lags were selected. We then estimate Eq. (1) using a Cochrane–Orcutt correction for autocorrelation of errors and compute tax revenue surprises $(T_t - T_t^*)/Y_t$. The size of the so-called French *cagnotte* in 1999 was between 50 and 80 billion Francs, i.e., between .7% and 1% of GDP. We chose a threshold of three fourths of a percentage point of GDP, $\bar{\tau} = .75\%$. Let us note that, unfortunately, we do not have the data to test if our selection criterium would select the 1999 French *cagnotte*.

This gives us a first list of episodes.⁴ To be as robust as possible to the choice of the forecasting model, we also perform this selection exercise using a VAR model including the growth rate of GDP, the growth rate of the GDP deflator, the consumption to GDP ratio, and a short run nominal interest rate. We also include a post-1973 dummy as an exogenous variable. For all the countries of the sample, we choose one lag and estimate the VAR over the longest sample available. It should

⁴ This list of episodes, as well as many extra tables, can be found in an extended version of this work, downloadable from <http://fpj.portier.free.fr/home>.

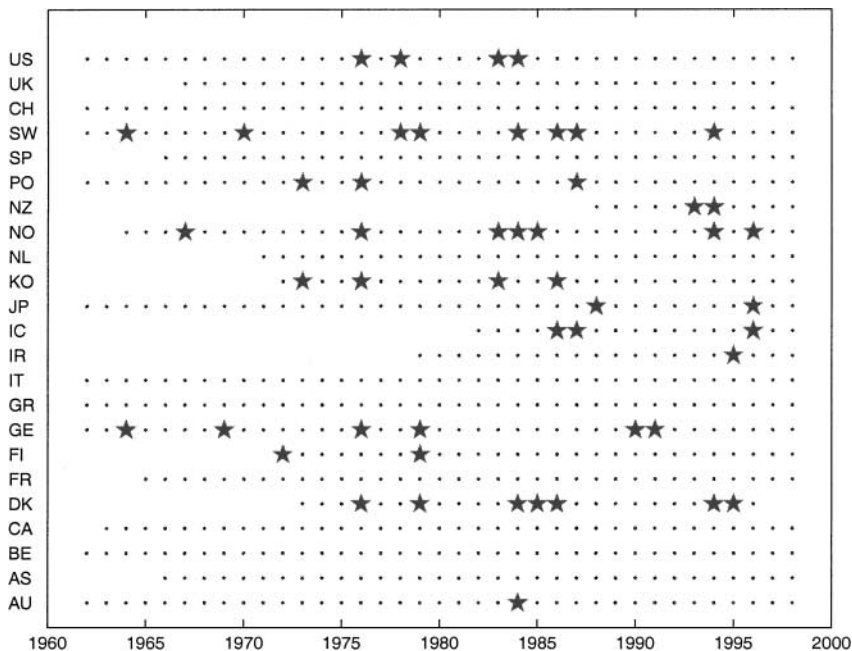


FIG. 1. Graphical representation of the selected episodes, linear tax schedule. On this figure are represented all the country-year episodes under consideration. A ★ represents a tax pot episode, a ● represents a non-tax pot episode, and a blank represents a country year in which data are not available to decide whether it is a tax pot episode. An episode is considered to be a tax pot if it is detected both by the AR and by the VAR model or if it is detected by the AR while the VAR model is not estimated for this country-year. The mnemonics correspondence is given in Table I.

be said that these samples are always smaller than those in the $AR(p)$ model. Equipped with this second forecasting model, we again select tax pot episodes for a .75% threshold. We have selected 68 episodes with the AR model and 63 with the VAR one. Of those episodes, 45 are common to the two selection device and four episodes in the early 1960s are chosen by the AR model, while data are not available for those country-years with the VAR model. Let us temporarily choose to keep the episodes that were selected by both methods plus the four episodes that cannot be detected by the VAR model. Figure 1 allows for a visual inspection of the selected episodes.

Before going deeper into the description of these episodes, we perform some more robustness checks of the selection device and will end up with a slightly modified list.

Some more on the robustness check. We have seen already that the lists of episodes selected by an AR or a VAR model were mostly similar, which is a first robustness property of the episode list we have chosen until now. We now examine the robustness of this choice to the introduction of nonlinearity of the tax schedule.

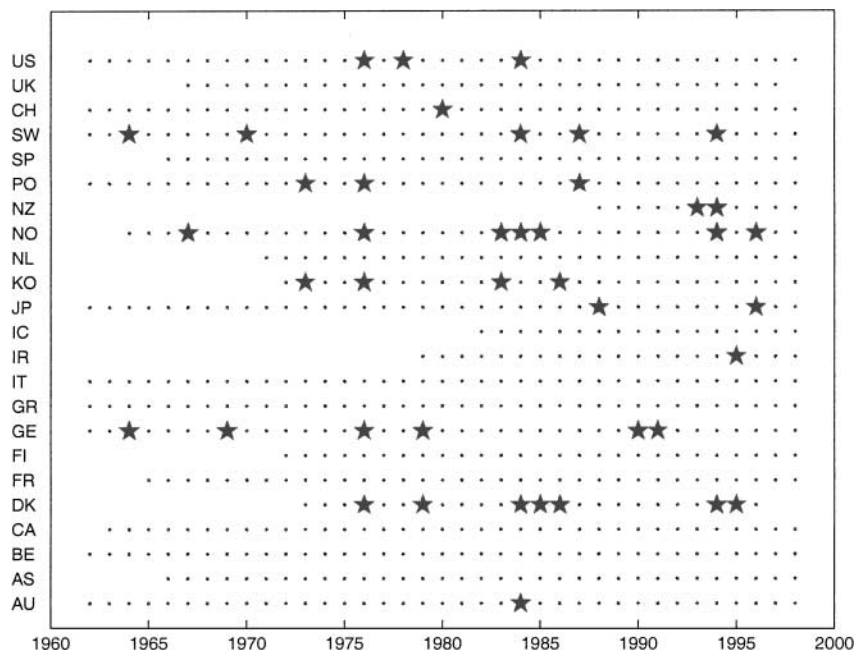


FIG. 2. Graphical representation of the selected episodes, progressive tax schedule. On this figure are represented all the country-year episodes under consideration. A ★ represents a tax pot episode, a ● represents a non-tax pot episode, and a blank represents a country year in which data are not available to decide whether it is a tax pot episode. An episode is considered to be a tax pot if it is detected both by the AR and by the VAR model or if it is detected by the AR while the VAR model is not estimated for this country-year. The mnemonics correspondence is given in Table I.

First, what happens to our list of tax pot episodes if we assume that the tax schedule is progressive and given by Eq. (7)? We conduct the AR and VAR predictive models for GDP again, but now compute T^* estimating (7) for each country. The episodes we select for the intersection of the AR and VAR models are displayed in Fig. 2.

Being more flexible, the progressive tax equation reduces the number of tax pot episodes selected from 50 to 42 when the intersections of AR and VAR models are considered. Furthermore, the selections obtained with the AR and with the VAR model are even closer: out of 50 and 54 for the two models, 42 are common. The nonlinear specification can be seen as a refinement of the linear tax model, as it eliminates some of the previous tax pot episodes, but does not add new ones, with the exception of Switzerland in 1980. This selection is the one we keep in the rest of the paper. The list of country-years that we eventually consider is therefore Australia 1984, Denmark 1976 1979 1984 1985 1986 1994 1995, Germany 1964 1969 1976 1979 1990 1991, Ireland 1995, Japan 1988 1996, Korea 1973 1976 1983 1986, Norway 1967 1976 1983 1984 1985 1994 1996, New Zealand 1993

1994, Portugal 1973 1976 1987, Sweden 1964 1970 1984 1987 1994, Switzerland 1980, and the United States 1976 1978 1984.

Note that adding past income in the tax equation changes the results only at the margin. Compared to the simple AR model, 54 episodes are selected as being tax pots (compared to 67), and 48 out of the 54 are in common with the simple AR model. In the following, we continue using the tax rule that does not include past GDP.

If we use a relative threshold definition of the tax pots, and we set κ to a level which allows for the selection of exactly 67 episodes (as in the simple AR model), this corresponds to the choice $\kappa = 1.19$. Out of those 67 episodes, 53 are in common with the simple AR model. We think that the intersection is large enough to make the results relatively insensitive to the choice of a relative or absolute threshold, and we decide to keep the absolute threshold definition in what follows.

Finally, we increase the threshold to 1 and 1.5% of GDP. In doing so, we reduce quite importantly the number of episodes. Out of 42 episodes with $\bar{\tau} = .75$, we keep only 23 with a 1% threshold and 11 with a 1.5% one. It is also relatively large countries that disappear when the selection is tighter (United States, Japan). On the contrary, decreasing the threshold to .5% of GDP increases the number of episodes to 77. In addition to being the size of the 1999 French *cagnotte*, we think that a .75% threshold satisfies the tradeoff we have between having enough episodes and being robust to specification errors by imposing a stringent enough criterion.

3.2. The Selected Episodes

Let us have now a closer look at the list of episodes we have selected that is displayed in Fig. 2. The total number of country-years we study was *a priori* 943 (23 countries times 41 years). Because of missing data, lags in models, etc., we ended up with 726 country-year data points. Out of those 726, 42 are identified as tax pot episodes, which represents 6% of the available data points. Roughly speaking, a country is expected to experience one tax pot episode every 20 years, or one country out of the 23 under study is expected to experience a tax pot every year. Of course, these are unconditional probabilities and the point of this paper is to inspect conditional probabilities.

Let us look at the country list: 11 countries never experienced a tax pot (Austria, Belgium, Canada, France, Finland, Greece, Italy, Iceland, Netherlands, Spain, and the United Kingdom). Four countries out of the Big 7 are in this list; 8 out of the 15 are members of the European Union. Twelve countries experienced at least one episode (Australia, Denmark, Germany, Ireland, Japan, Korea, Norway, New Zealand, Portugal, Sweden, Switzerland, and the United States). Strikingly, 19 out of the 42 tax pot episodes occurred in Nordic countries (Denmark (7 episodes), Norway (6 episodes), and Sweden (7 episodes)). We have no good explanation for this agglomeration in Northern Europe, but this is a fact that is robust to selection criteria. In particular, it is not an *artefact* of a higher average tax to GDP ratio,

TABLE II
Level and Variability of Tax Revenues

	Average level (% of GDP)	Variability (s.d./mean $\times$ 100)
Australia	26.31	14.79
Austria	42.53	8.32
Belgium	43.27	16.44
Canada	32.41	12.63
Denmark	46.58	7.21
Finland	42.14	11.15
France	42.32	10.28
Germany	41.14	9.01
Greece	26.00	21.66
Iceland	33.21	4.21
Ireland	29.98	6.17
Italy	34.96	20.17
Japan	24.58	18.22
Korea	19.17	16.00
Netherlands	44.60	6.07
New Zealand	38.94	2.40
Norway	41.80	12.25
Portugal	26.92	30.21
Spain	29.72	25.09
Sweden	48.40	16.24
Switzerland	37.22	16.79
United Kingdom	30.47	6.58
United States	29.42	5.74

Note. We compute for each country a measure of the size of total revenues (the mean of the ratio total tax revenues/GDP) and a measure of the volatility of this ratio (the mean of the ratio total tax revenues/GDP divided by the standard deviation of the same ratio). The definition of total revenues is given in the data appendix.

as a relative threshold would also lead to an overrepresentation of Nordic countries. Table II shows that there is no clear relation between the average tax over GDP ratio, its variability, and the selection of a country-episode. The French tax to GDP average ratio is as large as the Danish one (respectively 42.3 and 46.6%) and the French volatility of this ratio is higher (10.3 vs 7.2%), but nevertheless, we find eight country-years of tax pot in Denmark and none in France. Belgium and Japan both have a very large volatility of the tax/GDP ratio (respectively 16.4 and 18.2%) and Belgium's average tax/GDP ratio is almost twice the Japanese one (43.2 vs 24.6%), but no tax pot is found in Belgium, while two are observed in Japan.

The average year of a tax pot episode is 1984. We observe 4 tax pot episodes in the 1960s, 12 in the 1970s, 14 in the 1980s and 12 in the 1990s. Note that a lot of observations are missing for the 1960s (see Fig. 2). Once this fact is taken into account, it seems that episodes are more or less uniformly distributed across decades.

We now turn to the macroeconomic environment during those episodes.

3.3. Macroeconomic Environment

Table III compares some macroeconomic indicators for tax pot episodes and for the average episode. In this table, as in the following ones, we use for each indicator the largest sample we can get for each country. Therefore, the number of episodes under consideration is not exactly the same between variables. For example, the maximum number is 726 when we look at unemployment or growth and the minimum is 560 when we consider public debt variations. On the left-hand side of the table, we use raw data (levels). Note also that a third column reports the standard deviation of the statistics under consideration, as computed on all country-years. This allows for an informal test of the meaningfulness of the difference between tax pot episodes and average episodes. On the right-hand side of the table, we also compute tax pot statistics in deviation from the mean of the country (*within*) to get rid of possible country-specific fixed effects. Because it is not clear whether it is more meaningful to compute those statistics with or without the country fixed effect, we have kept both sets of statistics in the tables. In general, they both lead to the same qualitative conclusion. In the text, we comment on the “level” statistics, except when some more information is provided by the “within” ones.

Tax pots occur slightly later (the average year is 1984 *vs* 1982), but this might be mainly due to the fact that we cannot detect tax pots in the early 1960s

TABLE III
Macroeconomic Conditions

	Levels			Within	
	Tax pot episodes	All episodes	s.d.	Tax pot episodes	s.d.
Average year	1984.09	1982.59	—	1984.09	1982.59
$\Delta \log(\text{GDP})$	6.92	3.28	2.70	3.27	2.51
Relative deviation from HP trend	0.45	0.02	2.85	0.45	2.85
Relative deviation from HP trend in $N - 1$	-3.04	0.02	2.86	-3.05	2.86
Unemployment	3.98	4.02	2.74	0.80	2.21
$\Delta \log(\text{Unemployment})$	-12.50	5.06	24.93	-0.49	0.63
Inflation	4.63	6.55	5.49	-1.30	4.77
Long-term nominal interest rate	9.27	10.24	4.79	0.30	3.66
Debt/GDP	50.37	50.89	27.89	1.35	19.87
$\Delta \log(\text{debt})$	9.24	12.50	10.93	-2.18	9.88
$\Delta \log(\text{Debt/GDP})$	-2.37	2.83	9.75	-4.06	9.36
Primary Surplus/GDP	0.54	1.78	3.64	-0.80	3.04
$\Delta \text{Primary surplus/GDP}$	-0.93	-0.15	1.70	-0.78	1.69

Note. The first column displays the average of the variable considered over tax pot episodes, the second column the average over all the available country-years, and the third the standard deviation of this variable over all available country-years. The fourth column uses differences from the country average data. The fifth is the standard deviation over all country-years of the differences from the country average. Note that the number of observations may differ across lines, depending on data availability.

because of data availability reasons. The difference in GDP growth rates is striking, growth being twice as large during tax pots. This result comes almost by construction as we use GDP positive surprises in the detection of tax pot episodes. But the difference is large, suggesting that a large increase in activity is necessary for tax pot to happen. Not only is growth high during those episodes, but the economy is also above its long-run trend, as measured by Hodrick–Prescott trend (with smoothing parameter 400). Moreover, the economy is way below trend the year before, so that those episodes look like brusque rebounds of activity. On average, unemployment is not different during tax pot episodes, but its growth rate is of course very different: -12% during tax pot episodes vs 5% on average.

As far as nominal conditions are concerned, there are no clear differences between tax pots and the average, tax pots being slightly less inflationary, with long-term nominal interest rates slightly lower.

The average level of the public debt to GDP ratio is similar in both cases (50%). The second clear difference is the evolution of this ratio: -2.4 points in tax pots and $+2.8$ points on average. The evolution of this ratio is of course related to the evolutions of its numerator and denominator: as we saw before, GDP growth is higher during tax pots. But it is also the case that real debt growth is three points lower in tax pot episodes (roughly 9 vs 12%). Finally, the primary surplus to GDP ratio is three times lower in tax pots, despite the unexpected increase in revenues. Note also that this ratio strongly decreases in tax pot episodes (about -1 point).

To summarize, tax pots episodes are different from other episodes with respect to their growth (which is larger), their primary surplus (which is smaller), and the evolution of their public debt to GDP ratio (which is negative). Let us notice one other distinctive characteristic of tax pots: the economy the year before was significantly below its trend, while after it is slightly above. To use macro analysts' words, what matters is not a large growth of potential output, but a rapid filling in of Okun gaps.

We now turn to the detailed way in which public expenditures and revenues vary during those episodes and which actions appear to have reduced public debt or fostered growth.

4. HOW AND TO WHAT END?

We first examine the evolutions of public expenditures and receipts during tax pot episodes and compare them with the average country-year. We also examine the contributions of the different taxes to the tax pot. Then, given the large number of Nordic country-year episodes in the sample of tax pots, we examine whether there is a Nordic idiosyncrasy in the sample. Then we turn to the efficient use of the tax pot. Two criteria are considered: public debt reduction and growth fostering. In each case, we split the tax pot sample into two sets of country-years labeled “success” and “failure,” and look for differences between those two subsets. Eventually,

we try to relate the use of the tax pot (tax reduction, expenditures increase, debt reduction) to the initial conditions of the country-year.

4.1. *Evolution of Public Finance during Tax Pot Episodes*

Here we comment on the results presented in Tables IV and V. First, let us recall that the number of episodes is not constant across lines of these tables and that the variable “total revenues” is not constructed as the sum of the taxes and transfers that appear above in the tables (see the data appendix for an explanation). Therefore, one cannot get the variation of total taxes as the sum of all tax variations.

We observe that the ratio expenditures to GDP is decreased by .8% during tax pots while it is increased by .24% on average: during tax pots, expenditures increase less than GDP. Surprisingly at first sight, total public revenues do not increase, while they do on average. They even decrease relatively to the country mean (see the “within” column). This indicates that tax pots are episodes in which part of the windfall tax revenues are used to deliberately decrease taxes. Tax pots are periods of positive fiscal impulse, as computed using Blanchard’s methodology (Blanchard, 1993b): one observes a deliberated increase of the primary deficit to GDP ratio (what we call fiscal impulse) of 1.29%.

TABLE IV
Expenditures and Revenues

	Levels			Within	
	Tax pot episodes	All episodes	s.d.	Tax pot episodes	s.d.
Δ Total expenditures/GDP	−0.81	0.24	1.25	−1.04	1.23
Δ Total revenues/GDP	0.07	0.35	1.05	−0.23	1.04
Fiscal impulse/GDP	1.29	−0.02	0.73	1.31	0.73
Δ Gc/GDP	−0.64	−0.04	0.52	−0.61	0.50
Δ Gi/GDP	−0.03	−0.03	0.31	−0.03	0.30
Δ Gcw/GDP	−0.05	0.41	0.56	−0.37	0.52
Δ Tr/GDP	−0.20	0.30	0.71	−0.47	0.69
Δ Direct taxes/GDP	0.04	0.13	0.74	−0.00	0.73
Δ Indirect taxes/GDP	0.02	0.02	0.50	−0.03	0.49
Δ Personal income taxes/GDP	0.01	0.09	0.65	−0.11	0.65
Δ Corporate income taxes/GDP	0.16	0.05	0.42	0.14	0.41
Δ Employees social contributions/GDP	0.01	0.07	0.21	−0.06	0.21
Δ Employers social contributions/GDP	−0.01	0.09	0.41	−0.10	0.40
Δ Taxes on property/GDP	−0.01	0.00	0.22	−0.01	0.22
Δ Taxes on goods and services/GDP	0.07	0.03	0.51	0.01	0.50

Note. The first column displays the average of the variable considered over tax pot episodes, the second column the average over all the available country-years, and the third the standard deviation of this variable over all available country-years. The fifth is the standard deviation over all country-years of the differences from the country average. Note that the number of observations may differ across lines, depending on data availability. Note also that total revenues is not constructed as the sum of the revenues that are listed below in the table (see data appendix). Gi, government investment; Gc, government consumption; Gcw, wage government consumption; Tr, transfers from the government.

TABLE V
Tax Pot Components

	Tax pot episodes	All episodes	s.d.
Total revenues/GDP	1.22	-0.01	0.64
Transfers/GDP	-0.10	0.00	0.19
Personal income taxes/GDP	0.36	-0.00	0.22
Corporate income taxes/GDP	0.19	-0.00	0.13
Employees social contributions/GDP	0.06	-0.00	0.07
Employers social contributions/GDP	0.08	-0.00	0.09
Taxes on property/GDP	0.03	-0.00	0.07
Taxes on goods and services/GDP	0.30	-0.01	0.25

Note. The first column displays, for each variable and for all tax pot episodes, the average of the tax pot components, the second column the average over all the available country-years, and the third the standard deviation of these tax pot components over all available country-years. Note that the number of observations may differ across lines, depending on data availability. Note also that total revenues is not constructed as the sum of the revenues that are listed below in the table (see data appendix). The definition of tax pot components is given in Eq. (3).

As far as expenditures are concerned, government consumption to GDP ratio is reduced substantially. Transfers to GDP ratio also decrease, although less, while other expenditures stay approximatively constant in ratio.

As for revenues, we do not observe a substantial change in the personal income taxes, property taxes, and taxes on goods and services to the GDP ratio in tax pot episodes, while corporate income taxes increase more than on average and social contributions less than on average, again measured as percentage of GDP. Behind those apparent similarities, Table V shows that surprising and nondeliberated variations of taxes are indeed very different (they are by definition equal to zero on average over all episodes). The average nondeliberated and surprising increase in total revenues to GDP ratio is 1.22% for tax pot episodes, while it is by construction always greater than .75% of GDP. During a tax pot episode, personal income and taxes on goods and services contribute more to the surprising increase in tax revenues, followed by corporate personal income taxes. Other revenues are not very reactive as percentages of GDP.

To summarize, when we contrast tax pot episodes with the average country-year, we observe a decrease in the expenditure to GDP ratio, no increase in the revenues to output ratio, and a large positive fiscal impulse.

4.2. A Nordic Idiosyncrasy?

Before we study the efficiency of tax pot use to reduce debt and foster growth, let us inspect closer the fact that 19 of 42 episodes (45%) are Nordic country-years (Denmark, Sweden, and Norway). Here we split the tax pot episodes set into Nordic and non-Nordic countries to detect possible differences.⁵

⁵ The statistics relative to these two subsets can be found in the extended version of this work.

The average year is about the same for the two subsets. On average, growth is smaller in Nordic country-years of tax pots, the economy being less below trend the year before. Unemployment level and inflation are about the same, while the nominal interest rate is 200 basis points higher. Public debt level and evolution are roughly similar. One difference is that Nordic countries are running primary deficits during tax pots, while other countries have a primary surplus. The reduction in primary surplus to GDP ratio is twice as large in Nordic countries.

While expenditures and fiscal impulse are similar, total revenues to GDP ratio increases by almost 1 point in Nordic countries while it decreases by .2 points elsewhere. This is the main difference between the two sets of episodes, which is also observed using “within” statistics: there is no deliberate or nonsurprising decrease in taxes in Nordic countries at the time of a tax pot.

To summarize, the two subsets share more similarities than differences, and we treat them indistinctly in the following.

4.3. *How to Reduce Public Debt*

We now study under which conditions, if any, it had been possible to reduce public debt following a tax pot episode. We start from the entire set of tax pot episodes and compute for each one the variation of the public debt over GDP ratio between year N of the tax pot and year $N + 2$. If this variation is smaller than a negative threshold, we label this episode a success, while it is considered a failure if the variation is larger. We choose as a benchmark a threshold of -5% , which is such that of the 34 episodes we have (for the remaining 8, debt data are not available) 8 successes and 26 failures. Table VI presents the list of successes and failures. Note that in Denmark five tax pots are successes, while in Germany six are failure. Tables VII, VIII, and IX display the statistics we computed for both subsets of episodes.

Successes were not periods of larger growth, but unemployment decreased more than during failures. Inflation was also smaller, but we suspect this is a time fixed effect, as the average for successes is 1991, while it is 1982 for failures, a period of large inflation *ceteris paribus*. Debt was larger, but the debt to GDP ratio started immediately decreasing sharply (-6.7%), whereas it decreased ten times less for failures. The successful countries ran primary deficits the year of the tax pot, decreasing their primary surpluses by 190 GDP basis points. It seems that running larger primary deficits is the consequence of affecting more resources to debt reduction.

In terms of expenditures and revenues variations, let us first notice that the size of the tax to GDP ratio growth is twice as large for successes, while the tax pot is about the same size. The fiscal impulses are roughly of similar magnitude. Consequently, expenditures decreased twice more in successful episodes.

Are these conclusions robust to a higher threshold? With a -2% threshold, we now have 13 successes and 21 failures. The results are preserved: not larger

TABLE VI
Debt Reduction Criterion, 5% Threshold

	Success	Failure
Australia		
Austria		
Belgium		
Canada		
Denmark	1984 1985 1986 1994 1995	
Finland		
France		
Germany		1964 1969 1976 1979 1990 1991
Greece		
Iceland		
Ireland	1995	
Italy		
Japan		1988 1996
Korea		1973 1976 1983 1986
Netherlands		
New Zealand		
Norway	1994	1976 1983 1984 1985 1996
Portugal		1973 1976 1987
Spain		
Sweden	1987	1970 1984 1994
Switzerland		
United Kingdom		
United States		1976 1978 1984
Total	8	26

Note. The country-years of successful or nonsuccessful tax pot episodes are given. Here successful means that public debt was reduced by at least 5% the third year after the episode.

TABLE VII
Tax Pot Episodes, Debt Reduction Criterion, 5% Threshold

	Levels		Within	
	Success	Failure	Success	Failure
Average year	1991.20	1982.16	1991.20	1982.16
$\Delta \log(\text{GDP})$	5.78	6.68	2.41	2.70
Relative deviation from HP trend	0.15	1.02	0.15	1.02
Relative deviation from HP trend in $N - 1$	-2.42	-2.00	-2.42	-2.00
Unemployment	5.38	3.11	1.59	0.24
$\Delta \log(\text{Unemployment})$	-13.23	-6.94	-0.81	-0.26
Inflation	2.07	5.75	-3.85	-0.32
Long-term nominal interest rate	9.48	9.52	0.08	0.37
Debt/GDP	65.50	41.81	5.09	0.10
$\Delta \log(\text{debt})$	1.18	11.84	-7.90	-0.31
$\Delta \log(\text{Debt/GDP})$	-6.67	-0.62	-7.04	-2.81
Primary surplus/GDP	-0.13	0.81	-1.17	-0.26
$\Delta \text{primary surplus/GDP}$	-1.91	-0.99	-1.78	-0.87

Note. The first column displays the average of the variable considered over successful tax pot episodes, the second column the average over nonsuccessful ones. The third and fourth columns do the same thing for differences from the country average series (within). Here successful means that public debt was reduced by at least 5% the third year after the episode. Note that the number of observations may differ across lines, depending on data availability.

TABLE VIII
Tax Pot Episodes, Debt Reduction Criterion, 5% Threshold (*Continued*)

Expenditures and revenues	Levels		Within	
	Success	Failure	Success	Failure
Δ Total expenditures/GDP	-0.94	-0.47	-1.20	-0.78
Δ Total revenues/GDP	0.64	0.35	0.26	-0.01
Fiscal impulse/GDP	1.35	1.22	1.41	1.26
Δ Gc/GDP	-0.79	-0.55	-0.81	-0.49
Δ Gi/GDP	-0.06	0.00	-0.04	-0.02
Δ Gcw/GDP	-0.23	0.16	-0.62	-0.18
Δ Tr/GDP	-0.20	-0.02	-0.53	-0.29
Δ Direct taxes/GDP	0.33	0.14	0.24	0.04
Δ Indirect taxes/GDP	0.17	0.12	0.11	0.04
Δ Personal income taxes/GDP	0.14	-0.04	-0.01	-0.06
Δ Corporate income taxes/GDP	0.00	0.15	-0.05	0.12
Δ Employees social contributions/GDP	-0.02	0.06	-0.07	-0.02
Δ Employers social contributions/GDP	-0.01	-0.02	-0.13	-0.13
Δ Taxes on property/GDP	0.39	-0.03	0.39	-0.04
Δ Taxes on goods and services/GDP	0.33	0.06	0.25	0.01

Note. The first column displays the average of the variable considered over successful tax pot episodes, the second column the average over unsuccessful ones. The third and fourth columns do the same thing for differences from the country average series (within). Here successful means that public debt was reduced by at least 5% the third year after the episode. Note that the number of observations may differ across lines, depending on data availability. Note also that total revenues is not constructed as the sum of the revenues that are listed below in the table (see data appendix). Gi, government investment; Gc, government consumption; Gcw; wage government consumption; Tr, transfers from the government.

TABLE IX
Tax Pot Episodes, Debt Reduction Criterion, 5% Threshold (*Continued*)

Tax pot components	Success	Failure
Total revenues/GDP	1.27	1.17
Transfers/GDP	-0.07	-0.08
Personal income taxes/GDP	0.26	0.24
Corporate income taxes/GDP	0.14	0.16
Employees social contributions/GDP	-0.00	0.13
Employers social contributions/GDP	0.03	0.10
Taxes on property/GDP	0.04	-0.00
Taxes on goods and services/GDP	0.43	0.20

Note. The first column displays the average of the tax pot components over successful tax pot episodes, the second column the average over unsuccessful ones. Here successful means that public debt was reduced by at least 5% the third year after the episode. Note that the number of observations may differ across lines, depending on data availability. Note also that total revenues is not constructed as the sum of the revenues that are listed below in the table (see data appendix). The definition of tax pot components is given in Eq. (3).

growth, larger tax increase, similar tax pot, same fiscal impulse, larger decrease in expenditures. Those results are in line with Alesina and Perotti (1995).

To summarize, successes differ from failures not because the tax pot was larger, but because it was used differently. Country-years that increased primary deficits and affected the money-to-debt reduction effectively succeeded in reducing debt. Another minor difference is that the expenditures to GDP ratio decreases more in successes while the revenues to GDP ratio deliberately increases more.

4.4. *How to Foster Growth*

Let us repeat this exercise by now considering growth fostering as the criterion. To be precise, we compute for each country and for each year N the difference between its growth rate between N and $N + 2$ ($\gamma_{i,N}$) and the same growth rate of the Big 7 ($\gamma_{b7,N}$), that we denote $\hat{\gamma}_{i,N}$:

$$\hat{\gamma}_{i,N} = \gamma_{i,N} - \gamma_{b7,N}. \quad (9)$$

Let us denote by $\mu(\hat{\gamma}_i)$ and $\sigma(\hat{\gamma}_i)$ the mean and standard deviation of the series of deviations $\hat{\gamma}_{i,N}$ for a given country i . We then decide that an episode is successful in fostering growth when $\hat{\gamma}_{i,N} \geq \mu(\hat{\gamma}_i) + \kappa \times \sigma(\hat{\gamma}_i)$, where κ is a positive constant that will take the values 0 and 1. In words, in the case $\kappa = 0$, a success will be an episode such that within the next two years, the country performs relatively better than on average, as compared to the Big 7 growth rate. The criterion will be more strict with $\kappa = 1$.

Tables X to XIII present the results. Let us start with the looser criterion, i.e., $\kappa = 0$. We select 22 successes and 20 failures that are evenly distributed across countries. Successes happen on average later than failures. GDP growth is marginally larger in one subset the year of the tax pot, but in successful episodes, the country starts below its HP trend and was further below the year before. This could be enough to explain the success given the trend-reverting property of GDP.

Even though GDP growth is not larger, the unemployment decrease is (-15.7 vs -4.7%), although this difference is less spectacular when we correct for country fixed effect (*within*). The reduction of debt to GDP ratio is large for successes. It is also the case that the primary surplus to GDP ratio is larger for successes (1.5% primary surplus to GDP ratio) while failures experience primary deficits (.17%). Again, successes are episodes where the primary surplus decreases a lot, where most of the money is being used to decrease debt. As for debt criterion, success is a synonym of expenditures to GDP reduction, while this ratio is roughly constant for failures. Contrary to the debt criterion, the total revenues to GDP ratio is slightly decreased in successes, while it is increased in failures. Note that the size of the tax pot (nondeliberated and surprising variations in total revenues over GDP ratio) does not condition the success of an episode.

When we turn to $\kappa = 1$, we eliminate most of the episodes so that we end up with 5 successes for 37 failures. We basically draw the same conclusions. As before,

TABLE X
Growth Criterion, $\kappa = 0$

	Success	Failure
Australia	1984	
Austria		
Belgium		
Canada		
Denmark	1984 1994 1995	1976 1979 1985 1986
Finland		
France		
Germany	1969 1990	1964 1976 1979 1991
Greece		
Iceland		
Ireland	1995	
Italy		
Japan	1988	1996
Korea	1973 1976 1986	1983
Netherlands		
New Zealand	1993 1994	
Norway	1983 1984 1994 1996	1967 1976 1985
Portugal	1987	1973 1976
Spain		
Sweden	1994	1964 1970 1984 1987
Switzerland	1980	
United Kingdom		
United States	1976 1984	1978
Total	22	20

Note. The country-years of successful or nonsuccessful tax pot episodes are given. Here successful means that GDP growth was relatively higher the two years after the tax pot. The exact meaning of “relatively higher” and κ is given in the text (see Eq. (9)).

growth is not initially much larger, and it is still the case that the country is further below its HP trend for successes. Unemployment variations are now marginally larger for successes (within column), and we do not observe a larger reduction of the primary surplus. Tax pots are again of the same size, and again the total revenues to GDP ratio is decreased in successes, while it is increased in failures.

To summarize, it seems that growth will be high following a tax pot if the country was initially significantly below its trend, if expenditures are reduced, and if revenues are decreased, both as a percentage of GDP. The size of the tax pot is not related with the probability of success.

4.5. Understanding the Use of the Tax Pot

Let us consider three possible uses of the tax pot: decreasing taxes, increasing expenditures, or decreasing debt. Is it the case that highly indebted countries are more likely to choose debt reduction, highly taxed countries to reduce taxes, and high expenditures countries not to increase taxes? To answer those questions, we extract from the tax pot episode sample the set of high debt episodes (we also do

TABLE XI
Tax Pot Episodes, Growth Criterion, $\kappa = 0$

Macroeconomic conditions	Levels		Within	
	Success	Failure	Success	Failure
Average year	1986.63	1980.34	1986.63	1980.34
$\Delta \log(\text{GDP})$	6.66	6.50	3.02	2.67
Relative deviation from HP trend	-0.76	1.94	-0.76	1.94
Relative deviation from HP trend in $N - 1$	-4.11	-1.00	-4.11	-1.00
Unemployment	4.52	3.04	1.35	-0.07
$\Delta \log(\text{Unemployment})$	-15.70	-4.71	-0.64	-0.19
Inflation	4.35	5.25	-1.58	-0.80
Long-term nominal interest rate	9.76	9.20	0.79	-0.17
Debt/GDP	55.55	43.76	6.52	-1.08
$\Delta \log(\text{debt})$	7.32	12.47	-4.10	0.98
$\Delta \log(\text{Debt/GDP})$	-3.78	0.89	-5.47	-1.14
Primary surplus/GDP	1.50	-0.17	0.16	-1.06
$\Delta \text{ primary surplus/GDP}$	-1.13	-0.79	-0.98	-0.75

Note. The first column displays the average of the variable considered over successful tax pot episodes, the second column the average over nonsuccessful ones. The third and fourth columns do the same thing for differences from the country average series (within). Here successful means that GDP growth was relatively higher the two years after the tax pot. The exact meaning of “relatively higher” and κ is given in the text (see Eq. (9)).

the same for taxes and expenditures). A high debt episode is an episode in which the debt to GDP ratio is larger than the average ratio over all tax pot episodes plus one standard deviation of the same ratio over the same sample of tax pots. We then compute average variations of taxes, debt, and expenditures for the high group and for the whole sample of tax pots and repeat this exercise for tax and expenditure to GDP ratios. The results are gathered in Table XIV.

The average debt to GDP ratio is 50.37% for the 42 tax pot episodes, whereas it is 74.18% for the 11 episodes selected as having high debt. When we consider deviations from the country mean (*within*), we select 10 high debt episodes with an average deviation from the mean of 13.89%, whereas the deviation is 1.34% for all tax pots. The average tax revenue to GDP ratio is 34.58% for the 42 tax pot episodes, whereas it is 46.78% for the 21 episodes selected as having high taxes. When we consider deviations from the country mean (*within*), we select 15 high tax episodes with an average deviation from the mean of 4.02%, whereas the deviation is .38% for all tax pots. The average total expenditures to GDP ratio is 37.01% for the 42 tax pot episodes, whereas it is 49.83% for the 11 episodes selected as having high expenditures. When we consider deviations from the country mean (*within*), we select 9 high tax episodes with an average deviation from the mean of 4.51%, whereas the deviation is .32% for all tax pots.

What do we observe? High expenditures countries are likely to choose a smaller reduction of their debt, a smaller increase of their taxes, and a larger reduction

TABLE XII
Tax Pot Episodes, Growth Criterion, $\kappa = 0$ (Continued)

Expenditures and revenues	Levels		Within	
	Success	Failure	Success	Failure
Δ Total expenditures/GDP	-1.20	0.08	-1.43	-0.23
Δ Total revenues/GDP	-0.15	0.63	-0.44	0.26
Fiscal impulse/GDP	1.30	1.16	1.33	1.21
Δ Gc/GDP	-0.78	-0.38	-0.76	-0.34
Δ Gi/GDP	-0.05	0.08	-0.05	0.07
Δ Gcw/GDP	-0.07	0.14	-0.38	-0.21
Δ Tr/GDP	-0.23	0.14	-0.51	-0.18
Δ Direct taxes/GDP	-0.04	0.20	-0.09	0.10
Δ Indirect taxes/GDP	-0.11	0.26	-0.16	0.19
Δ Personal income taxes/GDP	-0.22	0.07	-0.14	-0.01
Δ Corporate income taxes/GDP	0.22	0.08	0.20	0.06
Δ Employees social contributions/GDP	0.03	0.05	-0.03	-0.02
Δ Employers social contributions/GDP	-0.09	0.09	-0.18	-0.01
Δ Taxes on property/GDP	-0.04	0.04	-0.04	0.03
Δ Taxes on goods and services/GDP	-0.08	0.31	-0.14	0.25

Note. The first column displays the average of the variable considered over successful tax pot episodes, the second column the average over unsuccessful ones. The third and fourth columns do the same thing for differences from the country average series (within). Here successful means that GDP growth was relatively higher the two years after the tax pot. The exact meaning of “relatively higher” and κ is given in the text (see Eq. (9)). Note that the number of observations may differ across lines, depending on data availability. Note also that total revenues is not constructed as the sum of the revenues that are listed below in the table (see data appendix). Gi, government investment; Gc, government consumption; Gcw, wage government consumption; Tr, transfers from the government.

TABLE XIII
Tax Pot Episodes, Growth Criterion, $\kappa = 0$ (Continued)

Tax pot components	Success	Failure
Total revenues/GDP	1.20	1.21
Transfers/GDP	-0.11	-0.06
Personal income taxes/GDP	0.37	0.25
Corporate income taxes/GDP	0.20	0.15
Employees social contributions/GDP	0.05	0.12
Employers social contributions/GDP	0.07	0.09
Taxes on property/GDP	0.03	-0.00
Taxes on goods and services/GDP	0.29	0.29

Note. The first column displays the average of the tax pot components over successful tax pot episodes, the second column the average over unsuccessful ones. Here successful means that GDP growth was relatively higher the two years after the tax pot. The exact meaning of “relatively higher” and κ is given in the text (see Eq. (9)). Note that the number of observations may differ across lines, depending on data availability. Note also that total revenues is not constructed as the sum of the revenues that are listed below in the table (see data appendix). The definition of tax pot components is given in Eq. (3).

TABLE XIV
Understanding the Use of the Tax Pot

	Levels				Within			
	All tax pots	High debt	High tax	High exp.	All tax pot	High debt	High tax	High exp.
Δ debt/GDP	-1.35	-2.56	-0.66	-1.77	-2.05	-2.18	-2.11	-1.71
Number of episodes	34	11	19	11	34	10	14	8
Δ tax/GDP	0.50	0.61	0.47	0.17	0.21	0.52	-0.03	-0.18
Number of episodes	42	11	21	11	42	10	15	9
Δ exp./GDP	-0.20	-0.72	-0.24	-0.74	-0.44	-0.27	-0.88	-1.11
Number of episodes	33	7	16	11	33	8	14	9

Note. We have extracted from the sample of tax pot episodes the ones with a high level of debt, taxes, or expenditures at the time of the tax pot (see the text for a precise definition of an episode with high debt, taxes, or expenditures). Then, we have computed the variation of debt, taxes, and expenditures the year of the tax pot, unconditionally and conditionally on being in a high episode. Those variations are taken in levels or in differences from the country average variation over the all time sample (within).

of their expenditures, which seems to be the primary goal. High tax countries choose a smaller increase of taxes and high debt countries a larger decrease of expenditures.

5. CONCLUSION

In this paper, we have proposed a methodology to select episodes in which tax revenues have been surprisingly and nondeliberately large, and have labeled those episodes tax pots. Using different criteria, we ended up with 42 country-year episodes among the 23 OECD countries and almost 40 years.

Tax pot episodes are different from other episodes with respect to their growth (which is larger), their primary surplus (which is smaller), and the evolution of their public debt to GDP ratio (which is negative). When we contrast tax pot episodes with the average country-year, we observe a decrease in the expenditure to GDP ratio, no increase in the revenues to output ratio, and a large positive fiscal impulse. Country-years with a high level of public expenditures are likely to choose a smaller reduction of their debt, a smaller increase of their taxes, and a larger reduction of their expenditures, compared to the average tax pot episode. Similarly, high tax countries choose a smaller increase of taxes.

Episodes that were followed by a significant decrease of the debt to GDP ratio did not receive a larger tax pot than the other country-years with a tax pot. Country-years that increased more primary deficits, reduced more expenditures, and increased more taxes are more likely to reduce their public debt the three years following the tax pot. Fostering growth is obtained when expenditures are reduced and if revenues are decreased, both as a percentage of GDP. The size of the tax pot is not related to the probability of success.

As in the fiscal adjustment literature, we then obtain relatively orthodox conclusions about the use of windfall tax revenues, as it is generally better for future growth and debt level to use the money to reduce expenditures and taxes.

APPENDIX A: DATA

All the data we use are taken from the OECD 2000 Statistical Compendium, and more specifically from National Accounts, Main Economic Indicators, Economic Outlook and Revenue Statistics.

A.1. Definitions

For Revenue Statistics, the OECD decomposes total tax revenues into six main subcategories: income and profit, social security, payroll, property, good and services, and others (with respective codes 1000, 2000, 3000, 4000, 5000, and 6000). When we decompose revenues, we only study the categories that are both large enough and homogenous enough across countries. We end up with a subdivision of 1000 into taxes on personal income (1100) and taxes on corporate income (1200), employees' social security contributions (2100), employers' social security contributions (2200), taxes on property (4000), and taxes on goods and services (5000). We have dropped payroll taxes (3000) (1% of OECD total tax revenue in 1997) and other taxes (6000) (3% of OECD total tax revenue in 1997). The total direct taxes (*TID*) and total indirect taxes (*TIND*) series are taken from Economic Outlook. Total revenues are constructed from Economic Outlook as direct taxes + indirect taxes + social contributions received by the government + other current transfers received by the government. Total transfers paid are computed as social benefits paid + other transfers paid by the government, again taken from Economic Outlook. Primary expenditures are the sum of total transfers paid, government consumption, public investment, and subsidies. The primary balance is then defined as primary expenditures – total revenues. Primary balance is constructed as total transfers paid – total revenues + government consumption + government investment + subsidies – government consumption of fixed capital – net capital transfers received by the government.

Fiscal impulse is constructed in a way similar to Blanchard (1993a) and Alesina and Perotti (1995). Cyclically adjusted total transfers paid and total revenues (the “star variables” defined in Section 2.2) are constructed and allow for the computation of a cyclically adjusted primary balance. The fiscal impulse is then defined as cyclically adjusted primary balance – actual primary balance. A positive value implies that some deliberated actions have been taken whose consequences are that actual surplus is lower than what it should have been, i.e., that there has been a positive fiscal impulse, not an adjustment.

GDP, unemployment, public debt, long-term nominal interest rate, aggregate consumption, and GDP price deflator are taken from Economic Outlook.

A.2. Exceptions

Because of data availability problems, there are some country-series that are computed in a different way. Those exceptions concern the computation of total transfers paid, total revenues, and primary balance. Note that the italicized uppercase mnemonics are referring to the Economic Outlook database.

- Total transfers paid: The rule is total transfers paid = social benefits paid + other transfers paid by the government. Exceptions are

United Kingdom: Total transfers paid = *GBRSSPG* + *GBRTRPG*

Denmark: Total transfers paid = *DNKTSUB* + *DNKSSPG*

Ireland, United States, Canada: Total transfers paid = social benefits paid

- Total revenues: The rule is total revenues = direct taxes + indirect taxes + social contributions received by the government + other current transfers received by the government. Exceptions are

Netherlands, Greece, United States: Total revenues = direct taxes + indirect taxes + social contributions received by the government

Ireland, Australia: Total revenues = direct taxes + indirect taxes + other current transfers received by the government

United Kingdom: Total revenues = personal income tax + corporate income tax + taxes on good and services + *GBRSSRG*

Denmark: Total revenues = personal income tax + corporate income tax + taxes on good and services + *DNKSSRG* + *DNKTRRG*

- Primary balance: The rule is primary balance = total transfers paid – total revenues + government consumption + government investment + subsidies – government consumption of fixed capital – net capital transfers received by the government. Exceptions are:

France, Portugal, Sweden: Primary balance = total transfers paid – total revenues + government consumption + government investment + subsidies – net capital transfers received by the government

Greece: Primary balance = total transfers paid – total revenues + government consumption + government investment + subsidies

United Kingdom: Primary balance = total transfers paid – total revenues + government consumption + government investment + *GBRTSUB* – *GBRKTRRG*

Denmark: Primary balance = total transfers paid – total revenues + government consumption + government investment + *DNKTSUB* – *DNKCFKG* – *DNKKTRRG*.

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